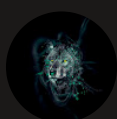




Connect a Web App to Amazon Aurora

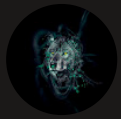


Kiprono Yegon

Sample page

NAME ADDRESS

ID	NAME	ADDRESS
1	Yegon	Nairobi
2	Pedri	Spain
3	Messi	Miami



Introducing Today's Project!

What is Amazon Aurora?

Amazon Aurora is a fully managed relational database service compatible with MySQL and PostgreSQL that combines the performance and availability of high-end commercial databases with the simplicity and cost efficiency of open-source databases. It is useful because it provides automatic backups, high availability, fault tolerance, fast failover, scalable storage, and read replicas for handling large workloads, while reducing the operational overhead of managing database infrastructure manually.

How I used Amazon Aurora in this project

In today's project, I used Amazon Aurora to store data that we served from the web page we had created

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the keenness needed when creating folders and the time it takes to debug a single error

This project took me...

It took me 45 minutes to complete the task

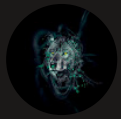
Creating a Web App

```
dan@y3gon:~/Devops/key-pairs$ ssh -i NextWorkAuroraApp.pem ec2-user@172.31.9.99
^C
dan@y3gon:~/Devops/key-pairs$ ssh -i NextWorkAuroraApp.pem ec2-user@35.90.177.150
The authenticity of host '35.90.177.150 (35.90.177.150)' can't be established.
ED25519 key fingerprint is SHA256:BJSxC6Zd79ot83AhWR/GwffdcQgoFUceiDQQei4nkis.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '35.90.177.150' (ED25519) to the list of known hosts
.

      #
    ~\  #####           Amazon Linux 2023
   ~~~~ \#####\
   ~~~~~ \###|
          \#/         https://aws.amazon.com/linux/amazon-linux-2023
            V~' '->
              .-.
             /  \
            /m/'  \
```

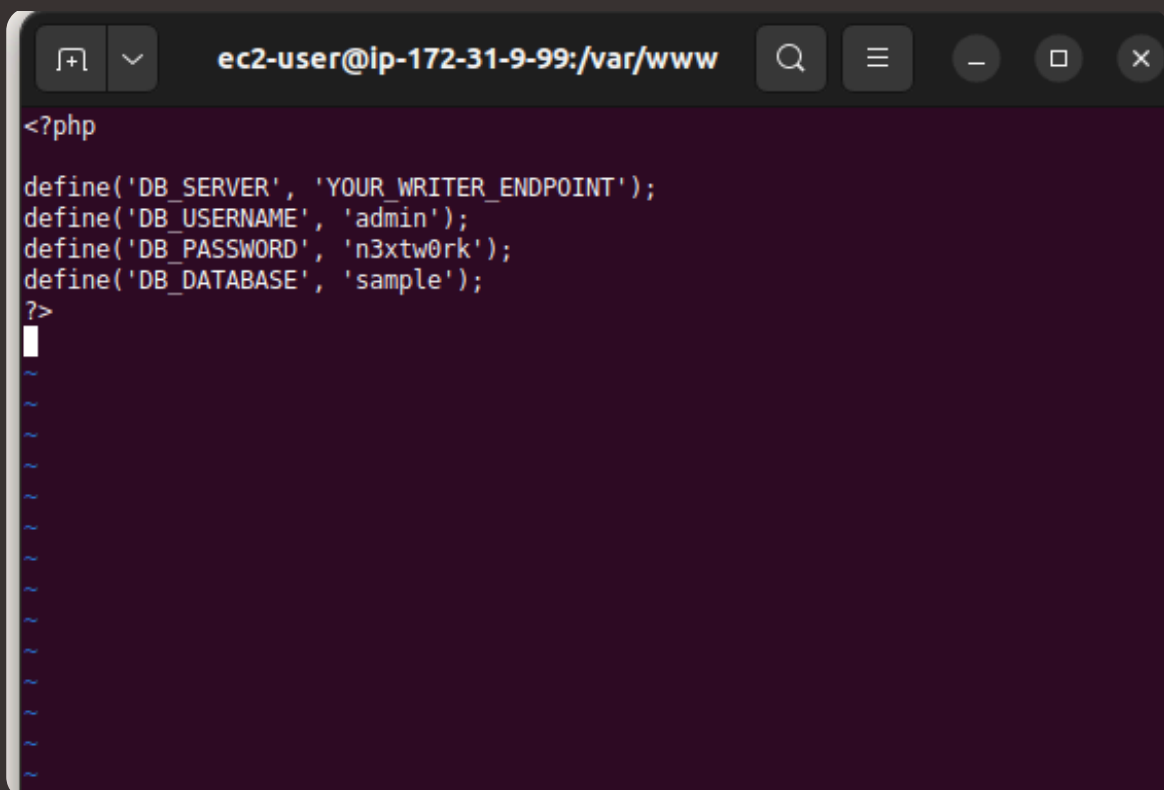
To connect to my EC2 instance, I ran the command `ssh -i NextWorkAuroraApp.pem ec2-user@35.90.177.150`

To help me create my web app, I first updated the instance and then installed the prerequisites to serve the web server: Apache web server(httpd), php, php-mysqli and mariadb105

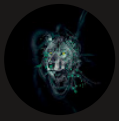


Connecting my Web App to Aurora

I set up my EC2 instance's connection details to my database by creating a file "dbinfo.inc" which contains the settings for the connection



```
<?php
define('DB_SERVER', 'YOUR_WRITER_ENDPOINT');
define('DB_USERNAME', 'admin');
define('DB_PASSWORD', 'n3xtw0rk');
define('DB_DATABASE', 'sample');
?>
```



My Web App Upgrade

Next, I upgraded my web app by creating a new file "SamplePage.php" and a ddinfo.inc to connect the database information to the web page

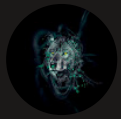
Sample page

NAME

ADDRESS

Add Data

ID	NAME	ADDRESS
----	------	---------



Testing my Web App

To make sure my web app was working correctly, I installed mySQL CLI and then checked the database if it contained the contents I had earlier added via the web page

```
ec2-user@ip-172-31-9-99:/var/www...  
  
mysql  
+-----+  
| performance_schema |  
| sample              |  
| sys                 |  
+-----+  
5 rows in set (0.001 sec)  
  
MySQL [(none)]> use sample;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
MySQL [sample]> show tables  
+-----+  
| Tables_in_sample |  
+-----+  
| EMPLOYEES         |  
+-----+  
1 row in set (0.002 sec)  
  
MySQL [sample]> desc EMPLOYEES;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra          |  
+-----+-----+-----+-----+-----+-----+  
| ID     | int unsigned  | NO   | PRI | NULL    | auto_increment |  
| NAME   | varchar(45)   | YES  |     | NULL    |                |  
| ADDRESS | varchar(90)   | YES  |     | NULL    |                |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.002 sec)  
  
MySQL [sample]> select * from EPMLLOYEES;  
ERROR 1146 (42S02): Table 'sample.EPMLLOYEES' doesn't exist  
MySQL [sample]> select * from EMPLOYEES;  
+-----+-----+-----+  
| ID | NAME | ADDRESS |  
+-----+-----+-----+  
| 1  | Yegon | Nairobi |  
| 2  | Pedri | Spain   |  
| 3  | Messi | Miami   |  
+-----+-----+-----+  
3 rows in set (0.001 sec)  
  
MySQL [sample]>
```



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